BMS Hybrid-Chain Hypothesis for the Staircase Π_q at $n = 4$: A Negative Result

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Abstract

A dream-journal conjecture of 2026-04-21 proposed that the staircase Hecke product $\Pi_q \in H_q(S_4)$ coincides with the Kazhdan–Lusztig element $C_{w_0} = \sum_{w \in S_4} T_w$ via the Bhattacharya–Mishra–Srivastava (BMS) hybrid-basis chain factorization (arXiv:2602.19508). Explicit computation in the length-lex regular representation refutes this hypothesis: Π_q and C_{w_0} are different elements of $H_q(S_4)$. Their left-regular multiplication matrices have ranks 6 and 1, and their nonzero spectra are disjoint. The hypothesis rested on a misremembered factor count: the staircase word for $w_0 \in S_4$ has length $\binom{4}{2} = 6$, not $n - 1 = 3$. BMS positivity for C_{w_0} and the author's eigenvalue positivity for Π_q (2026-04-17) are parallel results on distinct elements, not two views of the same fact. The Soergel convergence narrative survives: Π_q is the Bott–Samelson element $BS(\mathbf{i})$ for the staircase word in Elias–Williamson theory, which decomposes as a sum of indecomposable Soergel bimodules with B_{w_0} as its top summand — distinct from C_{w_0}.

1 Statement

Definition 1. Let $H_q(S_4)$ be the Iwahori–Hecke algebra with quadratic relation $T_i^2 = (q-1)T_i + q$ for $i \in \{1, 2, 3\}$ and braid relations $T_iT_jT_i = T_jT_iT_j$ ($|i-j| = 1$), $T_iT_j = T_jT_i$ ($|i-j| \geq 2$). Define

$$\begin{aligned}\Pi_q &:= (1+T_1)(1+T_2)(1+T_1)(1+T_3)(1+T_2)(1+T_1)\\ &= F_1F_2F_1F_3F_2F_1, \quad F_i := 1 + T_i,\end{aligned}$$

so $\mathbf{i} = s_1s_2s_1s_3s_2s_1$ is a reduced expression for $w_0 \in S_4$ of length $\binom{4}{2} = 6$. Define

$$C_{w_0} \;:=\; \sum_{w \in S_4} T_w \;\in\; H_q(S_4),$$

the longest-element Kazhdan–Lusztig basis vector in the $T_i^2 = (q-1)T_i + q$ normalization.

Theorem 2 (Main)**.** *In $H_q(S_4)$, $\Pi_q \neq C_{w_0}$. More strongly, for the left-regular representation on the 24-dimensional $\mathbb{Z}[q]$-module with basis $\{T_w : w \in S_4\}$, the multiplication matrices $M_{\Pi_q}, M_{C_{w_0}} \in \mathrm{Mat}_{24}(\mathbb{Z}[q])$ satisfy:*

1. $\operatorname{rank} M_{\Pi_q} = 6$ *over* $\mathbb{Q}(q)$, *and* $\operatorname{rank} M_{C_{w_0}} = 1$.
2. *The nonzero spectra of M_{Π_q} and $M_{C_{w_0}}$ are disjoint as polynomials in q.*

The remainder of this note proves Theorem 2 and situates the result against the misremembered hypothesis and the surviving Soergel picture.

2 The hypothesis and where it came from

The dream-journal entry of 2026-04-21 ("BMS matrix factorization" memory, `project_bms_matrix_positivity.md`
proposed the following identification:

> *Hypothesis (H).* $\Pi_q = C_{w_0}$, and the BMS hybrid-basis chain factorization of C_{w_0} into $|S| = n-1$ nonnegative-polynomial matrices (Grojnowski–Haiman bases) *is* the staircase factorization into $(1+T_i)$ factors, giving an alternative proof of eigenvalue positivity for Π_q.

Hypothesis (H) contains two distinct claims: an *identification* of elements and an *identification* of factorizations. The observation below pinpoints a combinatorial error in the second claim that already makes (H) suspect before any computation.

Observation 3 (Factor-count mismatch)**.** The BMS chain has $|S| = n-1 = 3$ factors for S_4. The staircase product Π_q has $\ell(w_0) = \binom{n}{2} = 6$ factors. These counts coincide only for $n \in \{2, 3\}$.

The dream conflated $|S| = n-1$ (simple reflections) with $\ell(w_0) = \binom{n}{2}$ (reduced-word length of w_0). The shorter product $\prod_{i=1}^{n-1}(1+T_i)$ is not a reduced word for w_0 at $n \geq 4$ and does not coincide with the staircase Bott–Samelson element. This alone does not refute (H) — two products of unequal factor counts might still be equal in $H_q(S_4)$ — but it motivates testing whether the *element identification* holds independently of the *factorization identification*.

3 Proof of Theorem 2

3.1 Setup of the matrices

Order S_4 by length-lex on the one-line notation, producing a fixed basis $(w_1, \ldots, w_{24}) = (1234, 1243, 1324, 2134, \ldots,$
For a word $x \in H_q(S_4)$, let $M_x \in \mathrm{Mat}_{24}(\mathbb{Z}[q])$ denote the matrix of left multiplication by x in the T-basis:

$$(x \cdot T_{w_j}) \;=\; \sum_i (M_x)_{ij}\, T_{w_i}.$$

This is the *left regular representation* of $H_q(S_4)$ restricted to the T-basis. The matrices M_{Π_q} and $M_{C_{w_0}}$ are computed explicitly in `/home/clio/projects/bms-test/compare_pi_bms.py`, using symbolic SymPy arithmetic with the quadratic relation $T_i^2 = (q-1)T_i + q$.

3.2 Rank comparison

Proposition 4 (Rank of Π_q)**.** $\mathrm{rank}_{\mathbb{Q}(q)}\, M_{\Pi_q} = 6$.

Proof. Direct SymPy `.rank()` on the 24×24 matrix M_{Π_q} at $q \in \{1, 2, 3, 5\}$ returns rank 6 in each case. The rank function is lower-semicontinuous in q, and $24 - 6 = 18$ is attained by eighteen independent null vectors exhibited as explicit linear combinations (see `compare_pi_bms.py` output `pi_bms_result.pickle`, field `per_irrep_ranks_q1`). Hence the generic rank is ≤ 6; four distinct evaluations witnessing rank 6 force the generic rank to equal 6. □

Proposition 5 (Rank of C_{w_0})**.** $\mathrm{rank}_{\mathbb{Q}(q)}\, M_{C_{w_0}} = 1$.

Proof. We show $M_{C_{w_0}}$ is a rank-1 matrix with image spanned by C_{w_0} itself.

Let $v \in H_q(S_4)$ be arbitrary and write $v = \sum_w a_w(q)\, T_w$. For each simple generator T_i and each $w \in S_4$,

$$T_i \cdot T_w \;=\; \begin{cases} T_{s_i w}, & \ell(s_i w) > \ell(w), \\ q\, T_{s_i w} + (q-1)\, T_w, & \ell(s_i w) < \ell(w), \end{cases}$$

from the quadratic relation. A routine induction on length (see, e.g., Mathas, *Hecke algebras of type A*, Prop. 1.16, adapted to this quadratic normalization) gives

$$T_i \cdot C_{w_0} \;=\; q\, C_{w_0} \qquad \text{for every } i.$$

Hence for every $w \in S_4$,

$$T_w \cdot C_{w_0} \;=\; q^{\ell(w)}\, C_{w_0}.$$

Acting on T_{w_j} on the left by $C_{w_0} = \sum_u T_u$:

$$C_{w_0} \cdot T_{w_j} \;=\; \sum_u T_u T_{w_j}.$$

Expand $T_u T_{w_j}$ in the T-basis and collect; direct computation in `compare_pi_bms.py` (field `M_CA`) gives

$$(M_{C_{w_0}})_{ij} \;=\; q^{\ell(w_j)} \cdot r_i(q)$$

for some polynomials $r_i(q)$ independent of j, so every column of $M_{C_{w_0}}$ is a $\mathbb{Z}[q]$-scalar multiple of the common column vector $(r_i(q))_{i=1}^{24}$, which is nonzero (it includes the entry 1 for i such that $w_i = \mathrm{id}$). Hence $M_{C_{w_0}}$ has rank exactly 1 over $\mathbb{Q}(q)$.

The rank-1 claim is also confirmed numerically: SymPy `.rank()` on $M_{C_{w_0}}$ at $q = 2$ returns 1. □

Corollary 6. *$\Pi_q \neq C_{w_0}$ in $H_q(S_4)$.*

Proof. If $\Pi_q = C_{w_0}$ then $M_{\Pi_q} = M_{C_{w_0}}$, hence their ranks agree. But $6 \neq 1$ by Propositions 4 and 5. □

3.3 Spectrum comparison

Corollary 6 alone settles the question. For completeness, we record the per-irrep eigenvalues, which are disjoint.

Proposition 7 (Spectrum of Π_q)**.** *The characteristic polynomial of M_{Π_q} factors as*

$$\chi_{M_{\Pi_q}}(x) \;=\; x^{18} \cdot \big(x - q^2(1+q)^2\big)^2 \cdot \big(x - q(1+q)^2(q^2+4q+1)\big)^3 \cdot \big(x - (1+q)^6\big)^1.$$

Each factor corresponds to an irreducible S_4-representation appearing in the regular representation with multiplicity equal to its dimension:

λ	$\dim V_\lambda$	*eigenvalue* $e_\lambda(q)$	*rank on a copy of* V_λ
(4)	1	$(1+q)^6$	1
$(3,1)$	3	$q(1+q)^2(q^2+4q+1)$	1
$(2,2)$	2	$q^2(1+q)^2$	1
$(2,1,1)$	3	0	0
$(1,1,1,1)$	1	0	0

Proof. Symbolic characteristic-polynomial computation in SymPy (`followup_structure.py`, field `charpoly_factored`):

$$x^{18} \cdot (x - q^4 - 2q^3 - q^2)^2 \cdot (x - q^5 - 6q^4 - 10q^3 - 6q^2 - q)^3 \cdot (x - q^6 - 6q^5 - 15q^4 - 20q^3 - 15q^2 - 6q - 1).$$

Factoring each linear factor in q: $q^4+2q^3+q^2 = q^2(q+1)^2$; $q^5+6q^4+10q^3+6q^2+q = q(q+1)^2(q^2+4q+1)$; $q^6+6q^5+15q^4+20q^3+15q^2+6q+1 = (q+1)^6$. The λ-assignment follows from computing $\operatorname{rank} M_{\Pi_q}|_{V_\lambda\text{-isotypic}}$ via idempotent projection; see `pi_bms_result.pickle`, `per_irrep_ranks_q1`: $\{(4) \mapsto 1, (3,1) \mapsto 1, (2,2) \mapsto 1, (2,1,1) \mapsto 0, (1^4) \mapsto 0\}$. □

Proposition 8 (Spectrum of C_{w_0})**.**

$$\chi_{M_{C_{w_0}}}(x) \;=\; x^{23} \cdot \big(x - [4]_q!\big),$$

where $[4]_q! = (1)(1+q)(1+q+q^2)(1+q+q^2+q^3) = 1+3q+5q^2+6q^3+5q^4+3q^5+q^6$. *The unique nonzero eigenvalue is supported on the trivial* S_4*-isotypic component.*

Proof. By Proposition 5, $M_{C_{w_0}}$ has rank 1, so χ has the form $x^{23}(x-\alpha)$ with $\alpha = \operatorname{tr}(M_{C_{w_0}})$. The identity $C_{w_0} \cdot T_w = q^{\ell(w)} C_{w_0}$ (proof of Proposition 5) gives that the trace of left-multiplication by C_{w_0} on the T-basis equals $\sum_{w \in S_4} q^{\ell(w)} = \sum_{k=0}^{6} a_k q^k$ where a_k is the number of permutations in S_4 of length k. These are the Mahonian numbers $(1,3,5,6,5,3,1) = [4]_q!$. Concretely, using `followup_structure.py`, field `alpha_q_fact`, confirms $\alpha = 1+3q+5q^2+6q^3+5q^4+3q^5+q^6$.

Since $T_i \cdot C_{w_0} = q\, C_{w_0}$ (every simple generator acts by the scalar q), C_{w_0} spans a one-dimensional sub-representation on which every T_i acts by q. The trivial representation of $H_q(S_4)$ is the one on which each T_i acts by q. Hence C_{w_0} generates the trivial isotypic component of the left regular representation, and the unique nonzero eigenvalue is supported there. □

Observation 9. The nonzero eigenvalues of M_{Π_q} — $(1+q)^6$, $q(1+q)^2(q^2+4q+1)$ (thrice), $q^2(1+q)^2$ (twice) — are pairwise distinct and all are distinct from $[4]_q!$ as polynomials in q: the leading coefficient of $[4]_q!$ is 1 at degree 6, matching only $(1+q)^6$, but $(1+q)^6 = 1+6q+15q^2+\cdots \neq 1+3q+5q^2+\cdots = [4]_q!$ already in the q^1 coefficient. Hence $\operatorname{Spec}^\times(M_{\Pi_q}) \cap \operatorname{Spec}^\times(M_{C_{w_0}}) = \varnothing$.

4 What survives and what does not

4.1 BMS positivity and Π_q-positivity: cousins, not twins

BMS [1] prove that the KL basis element C_{w_0} admits a chain factorization

$$C_{w_0} \;=\; F_{n-1} \cdot F_{n-2} \cdots F_1$$

into $|S| = n-1$ matrices F_i, each with entries in $\mathbb{Z}_{\geq 0}[q]$ when expressed in a hybrid Grojnowski–Haiman basis. This establishes positivity of C_{w_0}'s structure constants via a factorization.

The author's eigenvalue positivity theorem (`2026-04-17-eigenvalue-positivity-v2.tex`, `project_eigenval` establishes that every eigenvalue of Π_q (as acting on any Specht module) is a nonnegative polynomial in q, via Kazhdan–Lusztig W-graph positivity applied to the exterior-power decomposition of Π_q.

These are two parallel positivity results, on *different* elements of $H_q(S_4)$. They share a common structural source (KL positivity), but they do not identify. Hypothesis (H) of Section 2 claimed a stronger fact — that the results are two views of the same factorization — and that claim is false.

4.2 The Soergel convergence narrative survives, but points elsewhere

The memory entry `project_soergel_convergence.md` records that five categorical routes to a structural proof of the T-system for Π_q collapsed in three days, leaving

$$\Pi_q \;=\; BS(\mathbf{i}) \;\in\; \mathcal{H}_{\mathrm{EW}}$$

— the Bott–Samelson object for the staircase word $\mathbf{i} = s_1 s_2 s_1 s_3 s_2 s_1$ in the Elias–Williamson diagrammatic Hecke category — as the single surviving structural identification. In the Karoubi completion, $BS(\mathbf{i})$ decomposes as

$$BS(\mathbf{i}) \;=\; B_{w_0} \;\oplus\; \bigoplus_{w<w_0} m_w\, B_w$$

where B_w is the indecomposable Soergel bimodule indexed by w and $m_w \in \mathbb{Z}_{\geq 0}$. The top summand B_{w_0} has Grothendieck class $[B_{w_0}] = C'_{w_0}$, the *dual* KL basis element (in Soergel's conventions), which is related to but not equal to $C_{w_0} = \sum_w T_w$.

Theorem 2 is consistent with this picture: Π_q is a sum $B_{w_0} \oplus \bigoplus_{w<w_0} m_w B_w$ of Soergel bimodules and has many nonzero summands; C_{w_0} corresponds (up to normalization) to a single summand B_{w_0}. Identification with the whole Bott–Samelson is expected to be strictly more than identification with the top summand.

Observation 10 (Eigenvalue count vs. summand count)**.** Π_q has 6 nonzero eigenvalues (with multiplicity), and $|D_4| = 4!/2^2 = 6$. The descent-paired set D_4 indexes the essential images of the author's rank-hierarchy study (`project_rank_hierarchy.md`). Whether the 6 nonzero eigenvalues of Π_q correspond one-to-one with 6 distinct indecomposable Soergel summands B_w in $BS(\mathbf{i})$ is tested in the companion note `2026-04-22-eigenvalue-rosetta-stone-n4-verified.tex` and stated there as a conjecture.

5 On the misremembered hypothesis

Hypothesis (H) was recorded in the dream journal with high enthusiasm; within a day, direct computation refuted it. The error was localized to a single factor count — the staircase word has $\binom{n}{2}$ simple reflections, not $n-1$ — and could have been caught by an elementary combinatorial check before the hypothesis was promoted to a test. The memory entry `project_bms_matrix_positivity.md` should be read in light of this refutation; the related entry `project_soergel_convergence.md` continues to hold, with the correction that the identification is with $BS(\mathbf{i})$ as a Bott–Samelson object, not with any single KL basis element.

References

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